Assignment #7 constraints essay: TechniView

Team: Derek Corniello, Jason Bellerjeau, Ryan Sippy
Faculty advisor: Professor William Hawkins III

This revision describes the current course-linked coding practice platform and the planned February 15 to March 14, 2027 pilot. It replaces the earlier draft's interview-video and transcript assumptions. The ethics statement gives the detailed data, security, licensing, and AI-use record.

Economic

The team plans to host the pilot on University of Cincinnati resources rather than pay for managed application infrastructure. That choice requires the team to operate the application, MySQL, code runner, backups, and monitoring. The GPU host is within campus security and VPN controls; any optional local model for complexity estimates stays on that host. The old draft's $200-per-semester AWS cap is not a cost estimate for this UC-hosted pilot, and the current design does not record interview video or transcripts.

Security

Students submit source code and receive execution results tied to their course accounts. TechniView uses scrypt password hashes, expiring server-side sessions with HTTP-only cookies, course role checks, and a private code runner. The production configuration sets Secure cookies and Caddy HTTPS; the planned pilot runs within the campus VPN. The runner executes student processes with resource limits, seccomp, a separate network namespace, and reduced privileges on a dedicated host. The runner container itself is privileged, so host isolation, firewall rules, patching, and deployment checks remain necessary. UC-hosted encrypted storage and off-host encrypted backups protect stored submissions.

Legal

The pilot requires a UC IRB determination or approval before participant enrollment; approval is pending. Participants sign the pilot consent form before account creation. The team will follow applicable UC IRB requirements for retention, de-identification, access, and any FERPA obligations that apply to course records. The research export excludes names, emails, source code, raw output, and timestamps, but repeated pseudonymous codes and activity patterns can still reveal identity. The team reviews an export before sharing and removes the link between participant codes and account identities at pilot end.

Ethical

Participation is voluntary. Students can decline or withdraw without penalty; withdrawal stops new activity collection but does not erase data already collected, as the consent form explains. Participants at working sessions may receive free food. They keep food already received if they withdraw, and they need not finish problems to keep it. The team chose not to collect UC M-Numbers because internal account IDs are sufficient. Students see their own progress; authorized course staff and trained pilot team members have scoped access to identifiable results. The team will not treat a de-identified export as risk-free: repeated activity patterns can still make a participant recognizable.